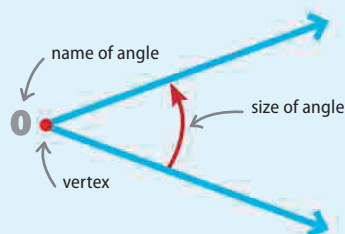
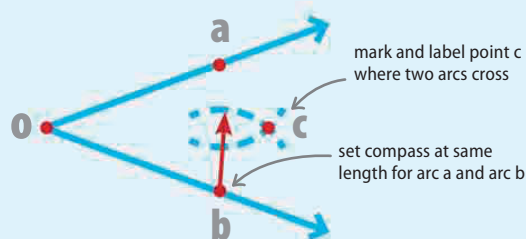


Bisecting an angle

The bisector of an angle is a straight line that intersects the vertex (point) of the angle, splitting it into two equal parts. This line can be constructed by using a compass to mark points on the sides of the angle.



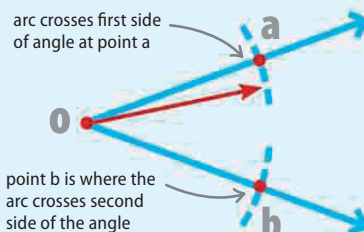
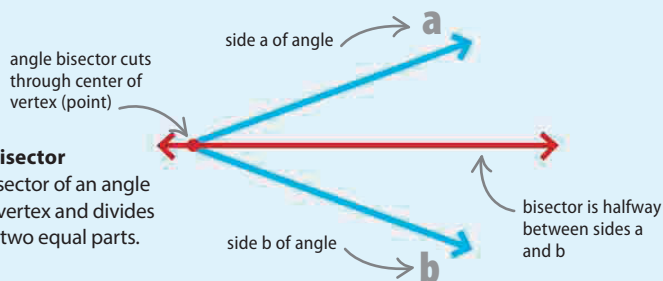
First, draw an angle of any size. Label the vertex of this angle with a letter, for example, o.



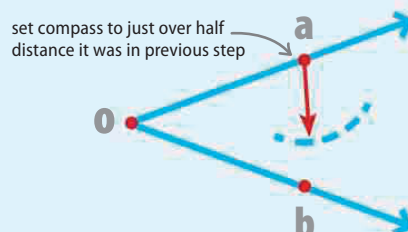
Keep the compass set at the same length and place it on point b, and draw another arc, and then on point a. The two arcs intersect at a point, c.

▷ An angle bisector

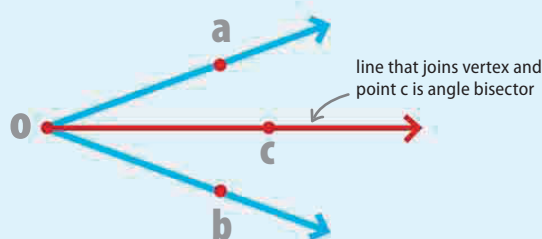
The interior bisector of an angle intersects the vertex and divides the angle into two equal parts.



Draw an arc by placing the point of a compass on the vertex. Mark the points at which the arc intersects the angle's sides and label them a and b.



Place the compass on point a and draw an arc in the space between the angle's sides.



Draw a line from the vertex, o, through point c—this is the angle bisector. The angle is now split into two equal parts.

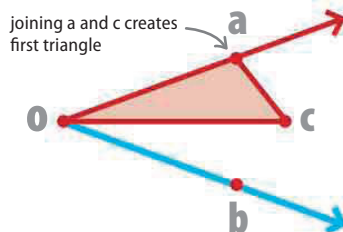
LOOKING CLOSER

Congruent triangles

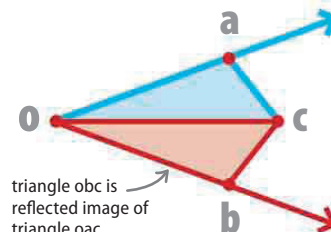
Triangles are congruent if all their sides and interior angles are equal. The points that are marked when drawing an angle bisector create two congruent triangles—one above the bisector and one below.

▷ Constructing triangles

By connecting the points made after drawing a bisecting line through an angle, two congruent triangles are formed.



Draw a line from a to c, to make the first triangle, which is shaded red here.



Now, draw a line from b to c to construct the second triangle—shaded red here.